

CYS 339: FRAUD AND DETECTION TECHNOLOGIES

Fraud and Detection Technologies

Mr. Mukombo Mukombo – mmukombo06@gmail.com

Who am I?

- Academician by training
  - Dip Ed
  - BICT
  - MSc IT

  Professional Affiliation
  - ICTAZ

  Research Interests:
  - Machine Learning
  - Internet of Things
  - Computer Security

Course Overview

- With today intensive use of integrated systems storing gigantic amount of financial and non financial data, hardly can any accounting, auditing or investigation work be conducted without data analysis. Valuable evidence may be staring at the accountant and auditor but not picked up if they do not know how to mine and connect the seemingly unrelated data to make sense out of it.

Course Overview

- The irreversible trend of Big Data and pressure to remain competitive have compel organization to make use of data analysis and continuous monitoring to understand business trend and consumer behaviours, as well as to detect fraud. Knowing what tool to use to acquire admissible digital evidence and how to search through pile of electronic files and email is a must have skill for today's auditors. This course will expose the learners to various data analysis techniques, tools and templates in the conduct of audit or investigation.

Course Objectives

- Acquire the overall concept of data analysis in detecting fraud.
- Better equipped to identify irregularities in data trend and data profile
- Able to perform qualitative data analysis.
- Understand rules governing digital evidence, digital forensic and online investigation.
- Understand about procedures in digital forensic and continuous monitoring.
- Gain knowledge on common detection techniques of fraud/ wrongdoing

Session Content

DATA ANALYSIS – OVERVIEW

- Definition of data analysis.
- Three Reason Data Analysis is Must.
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- Data Analysis in Auditing vs Investigation.
- What can Data Analysis do and can’t

What is data analysis?



What is data analysis?

- **Data analysis** is the process of systematically applying statistical and/or logical techniques to describe and illustrate, condense and recap, and evaluate data. It involves cleaning, transforming, and modeling data to discover useful information, draw conclusions, and support decision-making. (Sharma, 2019).

What is data analysis?

- **Data analysis** refers to the process of inspecting, cleansing, transforming, and modeling data with the goal of discovering meaningful patterns, extracting insights, and supporting decision-making processes. It often involves the use of specialized software and algorithms to process large datasets (Bryman & Bell, 2015)

What is data analysis?

- **Data analysis** is the science of examining raw data with the purpose of drawing conclusions about that information. It is used in various industries to enable organizations to make more informed decisions and verify or disprove existing models or theories (Tukey, 1977)

Other must know terminologies

- **Fraud** refers to the deliberate use of deception, dishonesty, or false representation to gain an unfair advantage, often resulting in financial or personal loss to the victim. In the context of cybersecurity, fraud typically involves unauthorized access to systems, data theft, or manipulation of information for malicious purposes. Examples include identity theft, phishing, credit card fraud, and insider threats (Anderson, 2020)

Other must know terminologies

Detection technologies are tools, systems, or methodologies used to identify and respond to malicious activities, anomalies, or threats within a network or system. These technologies are designed to monitor, analyze, and flag suspicious behavior, enabling organisations to mitigate risks and prevent cyber attacks. Examples include Intrusion Detection Systems (IDS), firewalls, machine learning algorithms, and anomaly detection systems. (Stallings & Brown, 2018)

Three Reasons Data Analysis is Must.



Why is data analysis a must?

Improved Efficiency in Detecting and Preventing Cyber Threats

- Data analysis enhances the efficiency of cybersecurity operations by automating the detection of anomalies and potential threats. Advanced analytical tools, such as Intrusion Detection Systems (IDS) and Security Information and Event Management (SIEM) systems, process vast amounts of data in real-time to identify suspicious behavior. This allows cybersecurity teams to respond quickly and effectively, minimizing downtime and preventing breaches **(**Stallings & Brown, 2018).

Why is data analysis a must?

Enhanced Understanding of Attack Patterns and Vulnerabilities

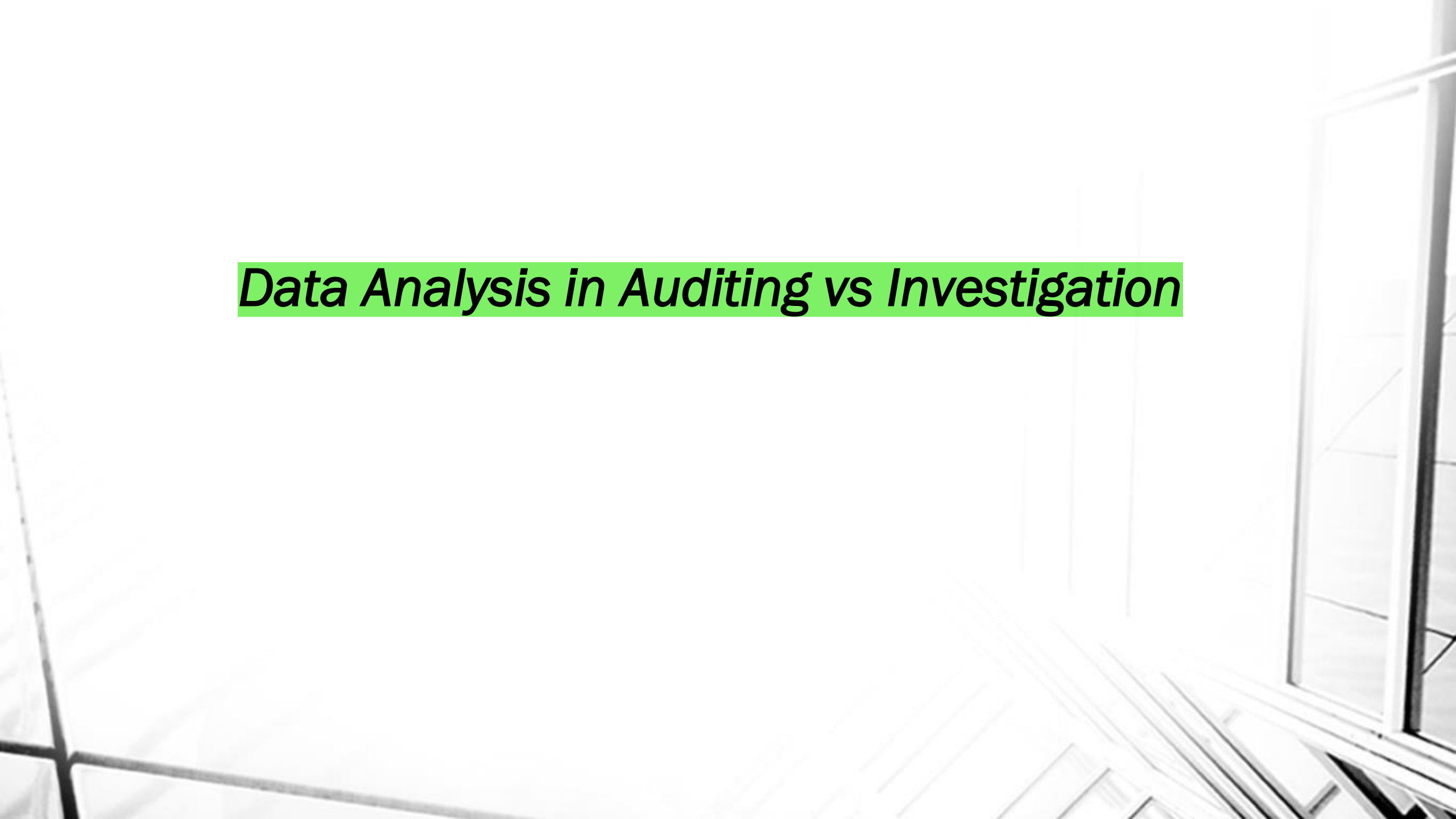
- Data analysis helps cybersecurity professionals gain deeper insights into attack patterns, tactics, and techniques used by malicious actors. By analyzing historical attack data and threat intelligence, organizations can identify vulnerabilities in their systems and develop strategies to strengthen their defenses. This proactive approach reduces the likelihood of successful cyberattacks and improves overall security posture. (Shostack, 2014)

Why is data analysis a must?

Informed Decision-Making for Threat Response

- Data analysis is critical in cybersecurity for making informed decisions about how to respond to threats and vulnerabilities. By analyzing data from logs, network traffic, and security incidents, cybersecurity professionals can identify patterns of malicious activity, assess risks, and prioritize responses. This enables organizations to take proactive measures to mitigate threats and reduce the impact of cyberattacks. (Anderson, 2020).

Data Analysis in Auditing vs Investigation



Data Analysis in Auditing

1. Purpose

- Data analysis in auditing is primarily used to assess the accuracy, completeness, and reliability of financial records and transactions.
- It helps auditors identify discrepancies, anomalies, or risks in financial statements.

2. Scope

- Focuses on evaluating internal controls, compliance with regulations, and the overall financial health of an organization.
- Often involves sampling and testing of data rather than examining every transaction.

Data Analysis in Auditing

3. Tools and Techniques

- Uses tools like Computer-Assisted Audit Techniques (CAATs) and software such as ACL or IDEA.
- Techniques include trend analysis, ratio analysis, and variance analysis.

4. Outcome

- Provides assurance to stakeholders (e.g., shareholders, regulators) that financial statements are free from material misstatements

(Kogan, Papanikolaou, Seru & Stoffman, 2017).

Data Analysis in Investigation

1. Purpose:

- Data analysis in investigations is used to uncover evidence of fraud, misconduct, or illegal activities.
- It aims to identify the root cause of issues and determine accountability.

2. Scope:

- Focuses on specific incidents or allegations, often requiring a deep dive into detailed transactional data.
- Involves examining all relevant data rather than relying on sampling

Data Analysis in Investigation

3. Tools and Techniques
- Uses forensic data analysis tools like EnCase, FTK, or Tableau for visualization.
- Techniques include link analysis, pattern recognition, and anomaly detection.

4. Outcome
- Provides actionable evidence for legal proceedings, disciplinary actions, or recovery of losses. (Singleton & Singleton, 2010)

Key differences between Auditing and Investigation

Objective

- Auditing focuses on providing assurance and compliance.
- Investigation focuses on uncovering wrongdoing and gathering evidence.

Approach

- Auditing is systematic and often follows a predefined plan.
- Investigation is reactive and driven by specific allegations or incidents.

Data Depth

- Auditing typically uses sampling and high-level analysis.
- Investigation requires exhaustive examination of detailed data.

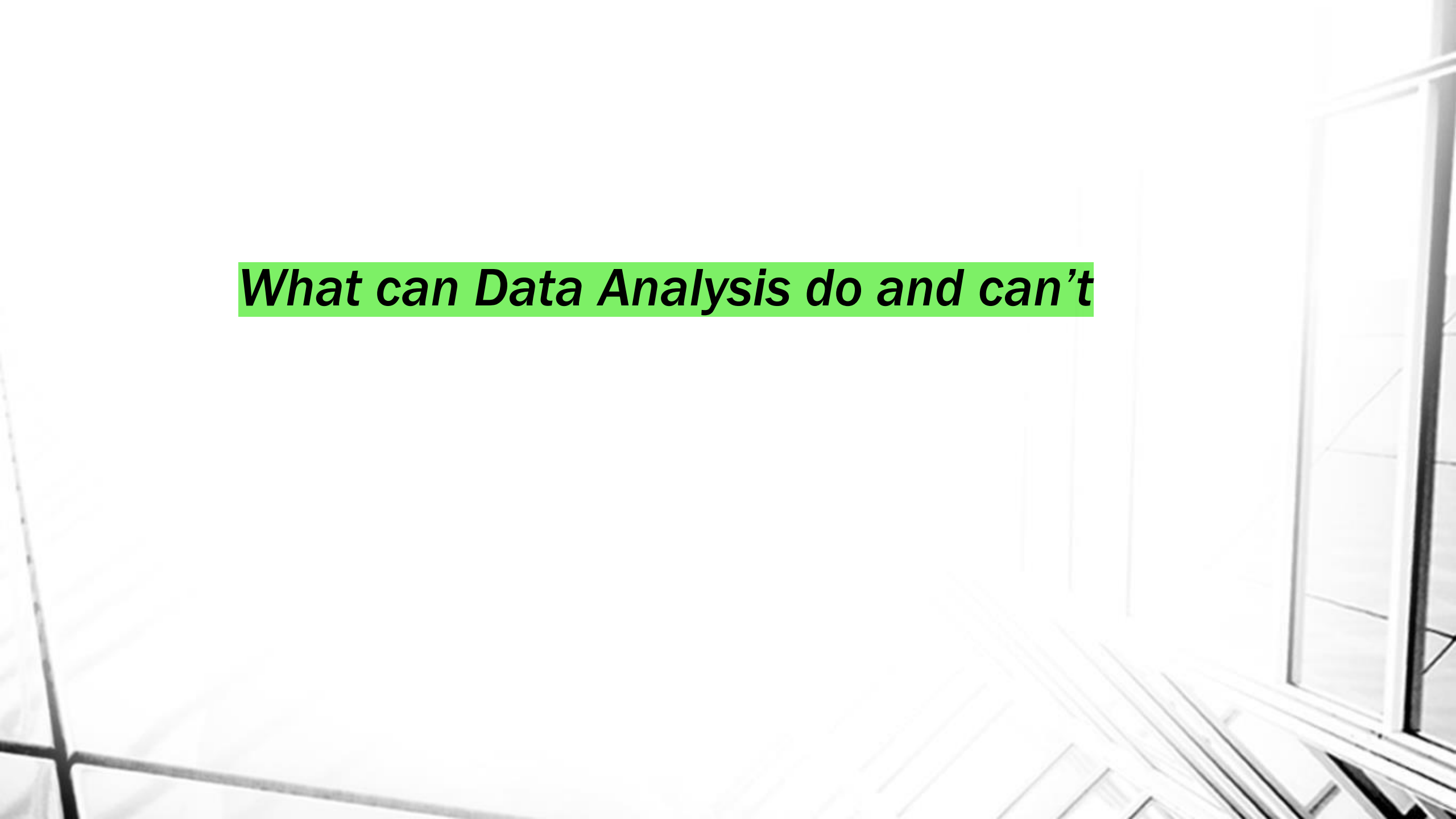
Key differences between Auditing and Investigation

Outcome Use

- Auditing results are used for stakeholder assurance and regulatory compliance.
- Investigation results are used for legal or disciplinary actions.

(Ramos, 2003).

What can Data Analysis do and can’t



What can Data Analysis do?

Identify Patterns and Trends in Cyber Threats:

- Data analysis can uncover patterns in network traffic, user behavior, or system logs to detect potential cyber threats, such as malware, phishing, or insider threats.

Example: Analyzing login attempts to identify brute force attacks or unusual access patterns that may indicate a compromised account. (Stallings & Brown, 2018).

What can Data Analysis do?

Support Decision-Making for Incident Response:

- Data analysis provides actionable insights to help cybersecurity teams prioritize and respond to incidents effectively.
- **Example:** Using data from Security Information and Event Management (SIEM) systems to determine the severity of an alert and allocate resources accordingly. (Anderson, 2020).

What can Data Analysis do?

Improve Efficiency in Threat Detection:

- By automating the analysis of large datasets, cybersecurity teams can detect threats faster and more accurately.

Example: Machine learning algorithms analyzing network traffic to flag anomalies that could indicate a Distributed Denial of Service (DDoS) attack. (Shostack, 2014).

What can Data Analysis do?

Detect Anomalies and Potential Breaches:

- Data analysis can identify deviations from normal behavior, such as unusual file access or data exfiltration, which may indicate a breach.

Example: Monitoring database queries to detect unauthorized access or SQL injection attempts.
(Ngai et al 2011).

What can Data Analysis do?

Enhance Threat Intelligence and Proactive Defense

- Data analysis helps organizations analyze threat intelligence feeds and historical attack data to predict and prevent future attacks.

Example: Correlating Indicators of Compromise (IoCs) from multiple sources to identify emerging threats.

(Kaspersky Lab. (2019).

What Data Analysis Cannot Do in Cybersecurity

Replace Human Judgment in Incident Response

- While data analysis can flag potential threats, it cannot replace the expertise of cybersecurity professionals in interpreting results and making decisions.

Example: A system may flag a legitimate user's unusual activity as suspicious, requiring human intervention to verify the alert. (Silver, 2012).

What Data Analysis Cannot Do in Cybersecurity

Guarantee Accuracy in Threat Detection

- Data analysis is only as good as the quality of the data and the algorithms used. False positives and false negatives can occur.

Example: A poorly tuned Intrusion Detection System (IDS) may miss sophisticated attacks or generate too many false alerts. (Few, 2012).

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